Smart glasses for binocular monitoring using high-speed off-axis microlens array camera

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Abstract

Continuous binocular monitoring provides emerging biomarkers for various neurodegenerative and ocular diseases. However, conventional gaze-tracking systems face limitations in daily healthcare monitoring due to their bulky form factors and insufficient imaging speeds. This work reports a smart glasses platform featuring a high-speed spatially-offset ellipsoidal microlens array (HS-SOEMLA) camera. The HS-SOEMLA was microfabricated via wafer-level processing and integrated onto a commercial CMOS image sensor. This ultrathin camera (total track length: 0.9 mm) is integrated into the center of the glasses frame to simultaneously capture both eyes at large off-axis angles. The ellipsoidal microlens curvatures, combined with spatially offset apertures, minimize off-axis aberrations at wide angles up to ±70°. Furthermore, periodic microlens alignment exploits the rolling-shutter read-out to segment the time domain, quintupling the native frame rate to 250 fps. This platform successfully tracks binocular oculometrics at high speed in wearable settings, facilitating continuous disease monitoring and early diagnosis in daily life.

Keywords: m*icrolens array camera, off-axis imaging, smart glasses, binocular oculometrics, healthcare monitoring*

1. Introduction

Early diagnosis improves the prognosis of neurodegenerative and retinal diseases, yet current clinical practices lack consistent screening methods [1]. Continuous binocular gaze tracking offers a promising approach to bridge this gap, but conventional systems impede daily use through bulky multi-camera architectures and limited imaging speeds [2]. This work introduces a smart glasses platform featuring a single high-speed spatially-offset ellipsoidal microlens array (HS-SOEMLA) camera (Fig 1). Mounted at the frame's center, the ultrathin camera captures both eyes simultaneously at steep off-axis angles. The ellipsoidal curvatures and spatially offset apertures suppress optical aberrations across an expansive field-of-view. Furthermore, a periodic microlens alignment exploits the rolling-shutter read-out to segment the time domain for high-speed binocular monitoring in a wearable format.

2. Results

2.1 HS-SOEMLA camera microfabrication & characterization

The HS-SOEMLA camera was monolithically microfabricated via wafer-level processing, utilizing Cr lift-off for apertures and thermal reflow for ellipsoidal microlenses (Fig. 2). Precise curvature control was achieved by adjusting microcylinder aspect ratios, with measured profiles showing a minimal 0.88% error from target designs. This camera achieves high-resolution, wide-angle imaging by employing angle-adaptive ellipsoidal microlenses with curvatures tailored to each incidence angle (Fig. 3). This spatially-varying design eliminates astigmatism and field curvature across the entire field-of-view. The system maintains superior spatial resolution and an infinite depth-of-field, even with an ultra-compact total track length of 0.9 mm. These high-resolution and depth-invariant properties facilitate close-range binocular monitoring without requiring complex optical assemblies.

2.2 High-speed binocular oculometrics in wearable settings

The fully integrated system validates the optical performance of the individual components through evaluation in a practical wearable scenario. The smart glasses prototype integrates the HS-SOEMLA camera with NIR LEDs into a wearable frame, powered by a portable battery and a Raspberry Pi 5 (Fig. 4). This platform captures high-contrast binocular images simultaneously, where the rolling-shutter-based reconstruction quintuples the frame rate to 250 fps (Fig. 5a). High-speed oculometrics successfully resolve rapid eye movements and blinks at a sub-10 ms timescale, revealing transient gaze position differences between eyes (Fig. 5b). Furthermore, the system precisely quantifies pupillary light reflexes and convergence changes during accommodation at varying depths (Fig. 5c). These results demonstrate the capability to track dynamic binocular biomarkers with high temporal resolution and accuracy in real-world wearable environments.

3. Conclusion

In conclusion, this work has successfully demonstrated a wearable smart glasses platform featuring an ultrathin HS-SOEMLA camera for high-speed binocular monitoring. This platform facilitates continuous and precise extraction of binocular biomarkers, providing a robust solution for the early diagnosis and continuous monitoring of neurodegenerative and retinal diseases.

Acknowledgments

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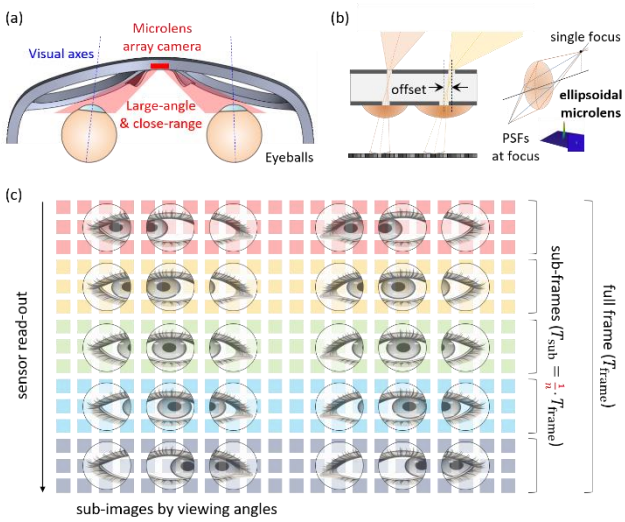

Fig. 1 (a) Smart-glasses platform for binocular eye monitoring. (b) Ultrathin microlens array camera that simultaneously captures both eyes while effectively suppressing optical aberrations at large off-axis angles. (c) Periodic microlens architecture facilitating high-speed binocular monitoring with a rolling-shutter image sensor.

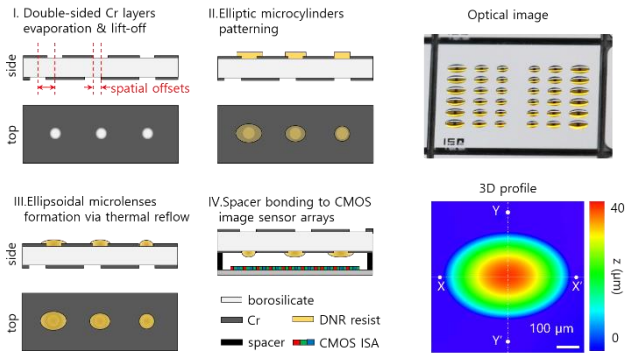

Fig. 2 Microfabrication steps and captures images of high-speed spatially-offset ellipsoidal microlens array (HS-SOEMLA).

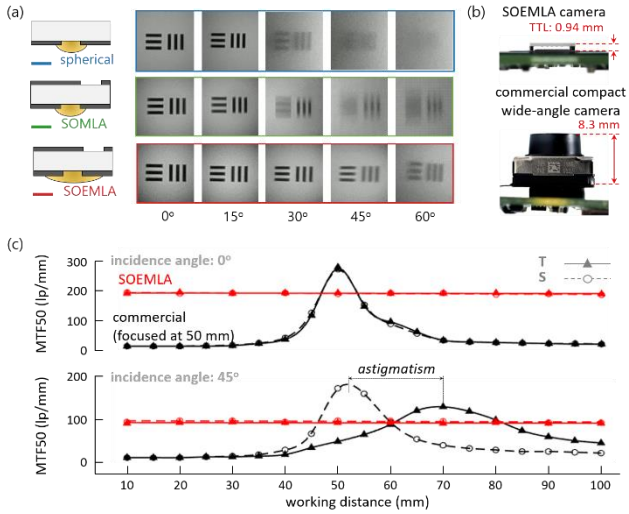

Fig. 3 (a) High-resolution imaging of the HS-SOEMLA camera maintained at large off-axis angles. (b) Ultrathin HS-SOEMLA camera design facilitating integration into smart-glasses frames. (c) Superior close-range imaging performance of the HS-SOEMLA camera compared with commercial wide-angle cameras.

Fig. 4 Smart-glasses prototype with an integrated camera and a side-view photograph of a subject wearing the device.

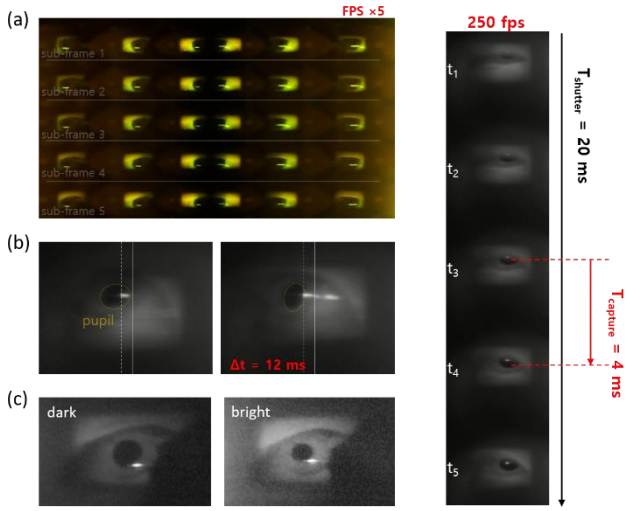

Fig. 5 (a) Captured image under ambient light (left) and a magnified image acquired under near-infrared illumination. A single image sensor is temporally multiplexed to acquire images at up to 250 fps (Δt = 4 ms). (b) Images showing pupil motion captured in a wearable configuration (Δt = 12 ms). (c) Images demonstrating pupil size variations in response to external conditions during wearable operation.